TOPSTECHNOLOGIES
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Windows Networking Services

❖ DHCP in Windows ServerTheory :How DHCP works: Scope, leases, and reservations.Configuring DHCP on a Windows Server.Practical :Set up a DHCP scope and allocate IP addresses.Task: Document the DHCP configuration with screenshots.

1. On your Windows Server VM, open the DHCP Manager and create a new DHCP scope named 'CampusWiFi' with an IP address range of 192.168.50.100 to 192.168.50.200 and a subnet mask of 255.255.255.0.

➢ Step 1: Open DHCP Manager

1. Log in to your Windows Server as Administrator.
2. Open Server Manager.
3. Click Tools → DHCP.
4. Expand your server name.
5. Right-click IPv4.
6. Select New Scope.

➢ Step 2: Configure scope name

1. Click Next.
2. Enter the Scope Name:
   CampusWiFi
3. Description (optional):
   DHCP scope for campus WiFi clients
4. Click Next.

➢ Step 3: Configure IP address range

Enter these details:

Field	Value
Start IP address	192.168.50.100
End IP address	192.168.50.200
Length	24
Subnet mask	255.255.255.0

➢ Step 4: Configure exclusions

For this assignment, leave the exclusion range empty unless you have specific addresses that must not be dynamically assigned.
Click Next.

➢ Step 5: Configure lease duration

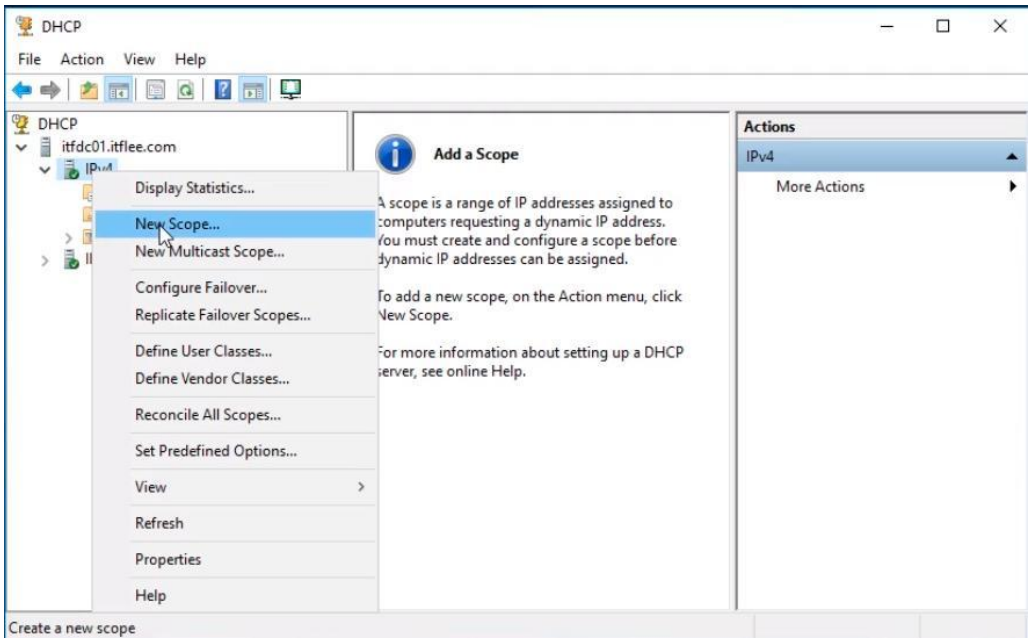
1. Set Days to 2.
2. Set Hours to 0.
3. Set Minutes to 0.

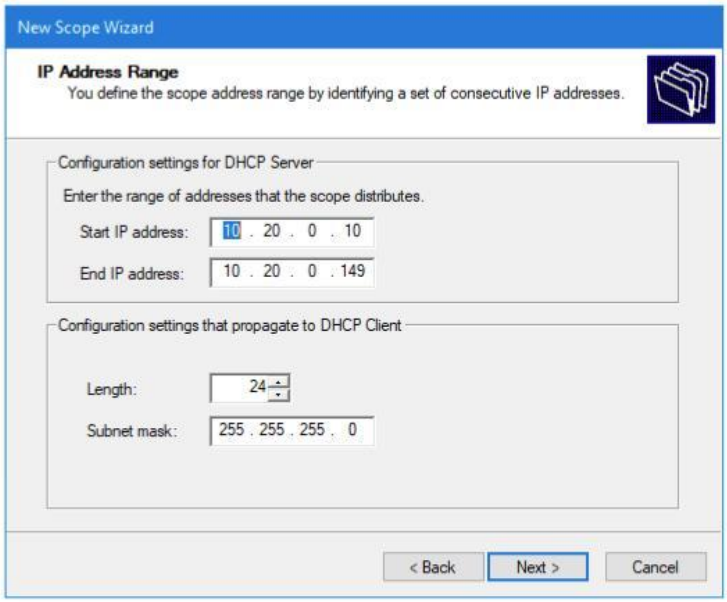
The resulting lease duration is:
2 Days 00 Hours 00 Minutes
Click Next.

- **Step 6: Configure DHCP options**

Continue through the wizard:

1. Select Yes, I want to configure these options now.
2. Configure the Router (Default Gateway) using your actual network gateway, if present.
3. Configure DNS Servers using your actual DNS server IP.
4. Skip WINS if it is not required.
5. Click Next → Finish.
6. If the scope is not yet active, right-click CampusWiFi and select Activate.





2. Add a DHCP reservation for a device (use your own laptop's MAC address if possible) so it always receives the IP address 192.168.50.150 from the 'CampusWiFi'scope. Hint: Right-click Reservations in your scope, choose 'New Reservation', and enter the MAC address.

- **Step 1: Find your laptop's MAC address**
  On your laptop, open Command Prompt and execute:
  ipconfig /all
  Find the active Wi-Fi or Ethernet adapter and note its Physical Address.
  Example:
  Wireless LAN adapter Wi-Fi:

  Physical Address: AA-BB-CC-DD-EE-FF
  This is an example MAC address. Use your actual laptop's MAC address.
  For a Windows laptop using Wi-Fi, check whether randomized hardware addresses are enabled for the network. Disable them for your test network when necessary, so the reservation continues matching the same MAC address.

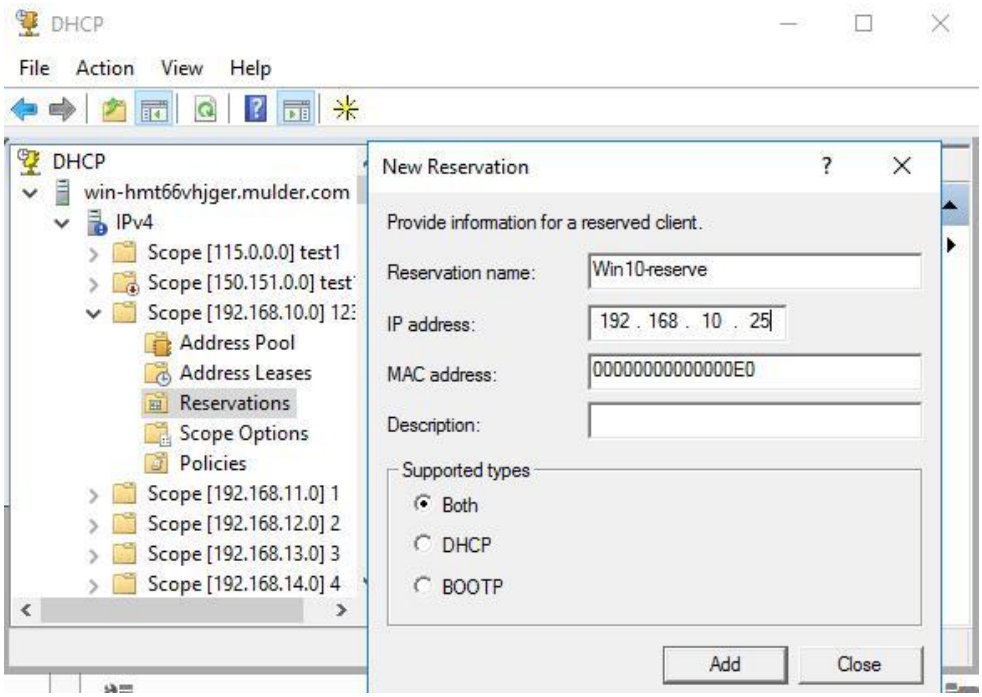
- **Step 2: Create the reservation**
  1. Return to your Windows Server.
  2. Open Server Manager → Tools → DHCP.
  3. Expand:
     DHCP
     └── Your Server
     └── IPv4
     └── CampusWiFi
     └── Reservations
  4. Right-click Reservations.
  5. Select New Reservation.

  Enter the following details:

Field	Value
Reservation name	Laptop-Het
IP address	192.168.50.150
MAC address	Your laptop's actual MAC
Supported types	DHCP

- **Step 3: Verify reservation**
  Expand Reservations and check that the entry appears:

Reservation name	Reserved IP
Laptop-Het	192.168.50.150

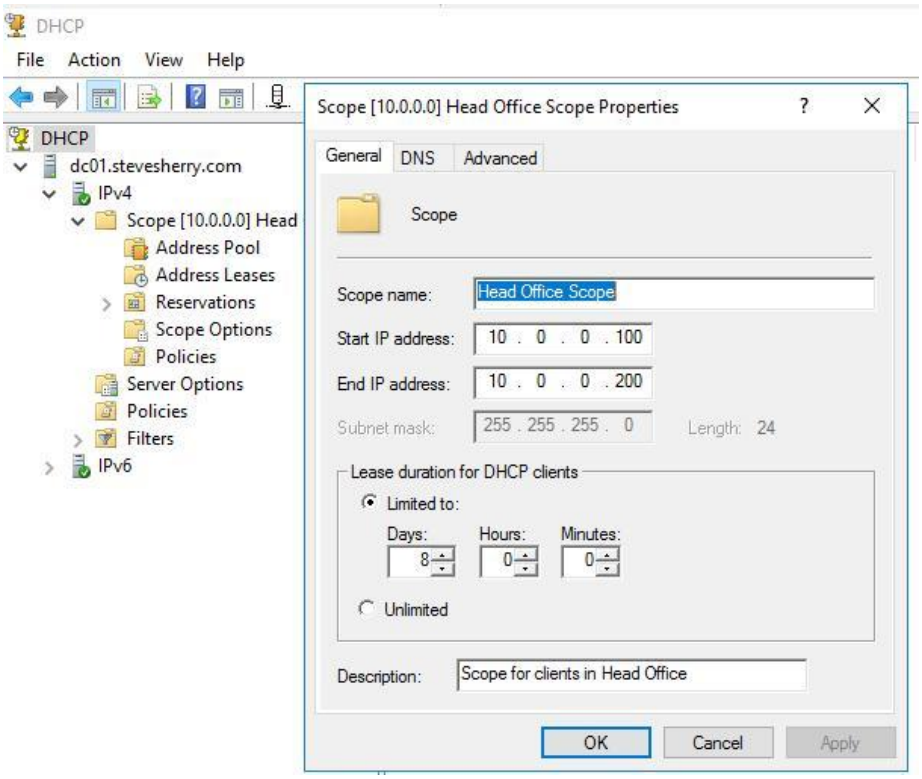


3. Configure the DHCP lease duration for the 'CampusWiFi' scope to 2 days instead of the default setting.

   We already configured two days during scope creation. Now verify and update it separately.

   - Steps:
     1. Open DHCP Manager.
     2. Expand IPv4.
     3. Right-click CampusWiFi.
     4. Select Properties.
     5. Open the General tab.
     6. Find Lease duration for DHCP clients.
     7. Enter:

Field	Value
Days	2
Hours	0
Minutes	0



4. Connect a client machine (or use a virtual client) to the network and verify it receives an IP address from your DHCP scope; take a screenshot showing the assigned IP in the client's network settings.

- **Step 1: Configure client network settings**
  On your client:
  1. Open Control Panel → Network and Internet → Network Connections.
  2. Right-click your active Ethernet or Wi-Fi adapter.
  3. Select Properties.
  4. Open Internet Protocol Version 4 (TCP/IPv4).
  5. Enable:
     - Obtain an IP address automatically.
     - Obtain DNS server address automatically.
  6. Click OK.

- **Step 2: Renew the IP address**
  Open Command Prompt as Administrator and execute these commands one by one:
  ipconfig /release
  ipconfig /renew
  ipconfig /all

- **Step 3: Verify the assigned IP**
  Since you created a reservation for this client's MAC address, the expected result is:

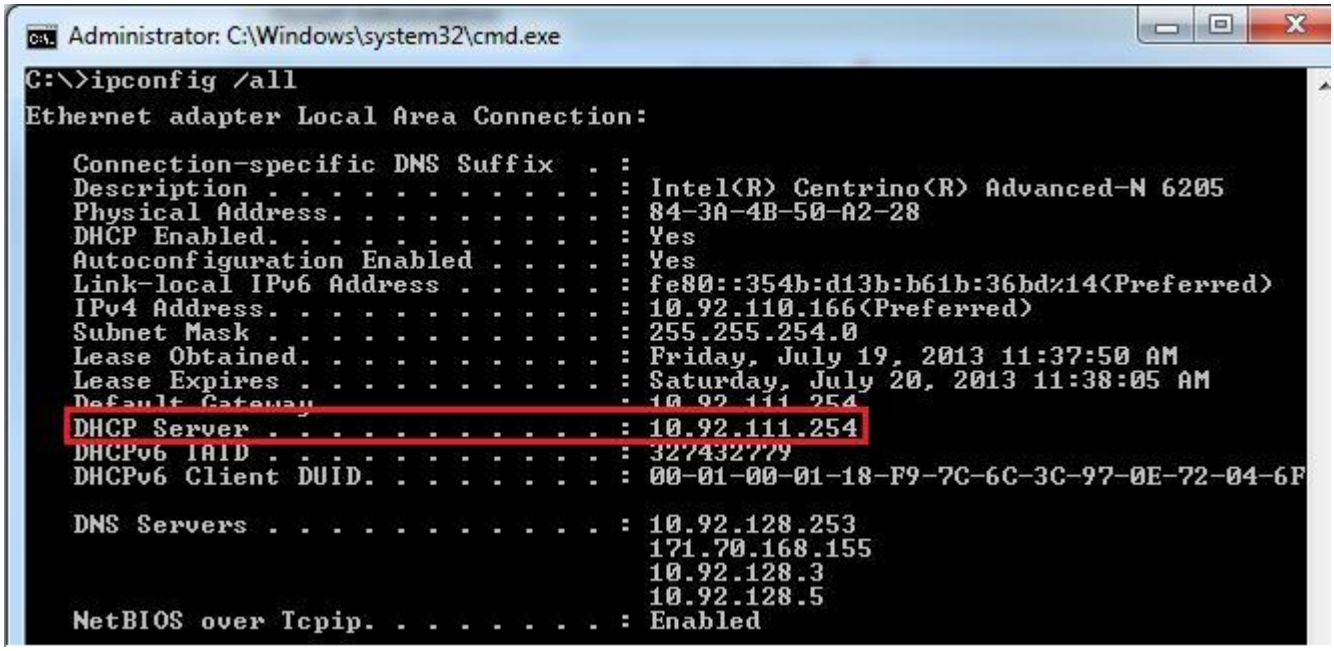
- IPV4 ADDRESS
  192.168.50.150

- Subnet Mask
  255.255.255.0

- DHCP Server
  Your server IP

- Lease Duration
  2 Days

- DHCP Enabled



➢ Important troubleshooting:

Problem	**Solution**
Client gets 169.254.x.x	Check DHCP scope activation and network connectivity.
Client receives a different subnet	Check VMware NAT or another DHCP server.
Client receives .100–.200 but not .150	Verify MAC address and reservation.
Client gets no gateway	Configure the correct Router scope option.
Client does not receive an address	Ensure server and client are on the same VLAN or configure DHCP relay.

5. Document your entire DHCP setup process in a Word or PDF file, including screenshots of the scope creation, reservation, lease duration change, and the client receiving its IP address.

   - You need to submit four actual screenshots with your practical report.

   - I have prepared a Word report containing the complete procedure, network configuration, commands, observation table, troubleshooting and screenshot placeholders.

   - **Screenshots required in your report**

Screenshot	What to capture
1	DHCP Manager showing CampusWiFi and IP range
2	Reservations showing 192.168.50.150 and MAC address
3	Scope Properties showing 2-day lease duration
4	Client ipconfig /all showing assigned IP address

   - The Word document has spaces to insert these screenshots. Replace those placeholders with your actual screenshots before submitting.